



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY,
DESIGN AND MANUFACTURING,
KANCHEEPURAM



Design of Gears and Transmission in BAJA ATVs: How Engine Types Influence the System



**Society of Automotive
Engineers - SAE Collegiate Club**

Presented by:
Nikil Narayana M
Powertrain Lead

Motivation?

Ever Wondered about these questions...?

- ? Why do BAJA ATVs need custom gear and transmission systems?
- ? How does a petrol engine behave in a rugged off-road race?
- ? How does an electric motor behave in the same race?
- ? Can we use the same transmission for both engines?

Let's find the answers for all these questions in this session.



Prerequisites

✓ Basic Understanding Of:

- Internal combustion engines – torque curves, throttle response
- Electric motors – BLDC/PMSM basics, instant torque behaviour

✓ Prior Exposure To:

- Gear ratios – how torque and speed trade-off with different ratios
- Torque-speed relationship – reading and interpreting performance curves
- CVT mechanisms – pulley-based variable transmission logic
- Chain drives – sprocket sizing and power transfer basics

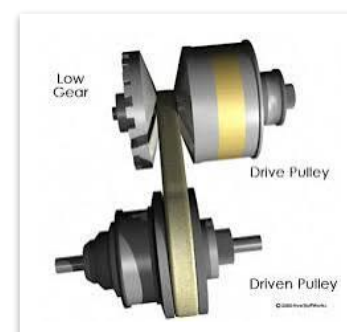


If you've worked on a Go-Kart, BAJA, or e-bike before — you're good to go!

Gear and Transmission Systems in BAJA ATVs

m-BAJA (IC Engine):

- Common setup: CVT + Chain-Sprocket Final Drive
- CVT enables smooth, automatic gear ratio variation for optimum torque
- Easy tuning for performance events like acceleration & hill climb



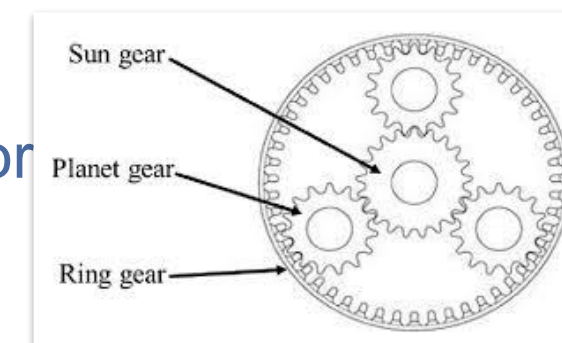
Manual Gearboxes:

- Rare in BAJA due to high driver effort and shifting complexity
- Increases weight and requires precise handling — not ideal for



Electric BAJA (eBAJA):

- Uses Planetary or Fixed Gear Reduction
- Instant torque from motor eliminates need for CVT
- More compact and efficient drivetrain layout



Final Drive – Chain & Sprocket:

- Used in both petrol and electric setups
- Simple, lightweight, and easily adjustable for changing gear ratios
- Ideal for rapid prototyping and tuning during testing



How Engine Type Changes Transmission Design

Feature	Petrol Engine (10HP)	Electric Motor (3–5kW)
Torque Curve	Mid RPM Peak	Instant at 0 RPM
Common Drive	CVT + Chain	Planetary + Chain
Complexity	Medium	Low
Maintenance	Medium	Low

! Key Insight:

Using the wrong transmission for a given engine leads to power loss and poor performance.
(E.g., CVT on an electric motor = unnecessary complexity; fixed gear on a petrol = sluggish pickup.)

Design Focus per Engine Type

Petrol Engine :

- Requires torque multiplication through a CVT to compensate for low-end torque
- Optimal performance when RPM is maintained between 2500–3000 (torque peak zone)
- CVT tuning (weights, springs) is crucial for different event types (acceleration vs endurance)
- Final drive achieved using chain-sprocket reduction (e.g., 10:1 or 12:1 total ratio)
- Needs regular CVT and belt maintenance in dusty off-road conditions



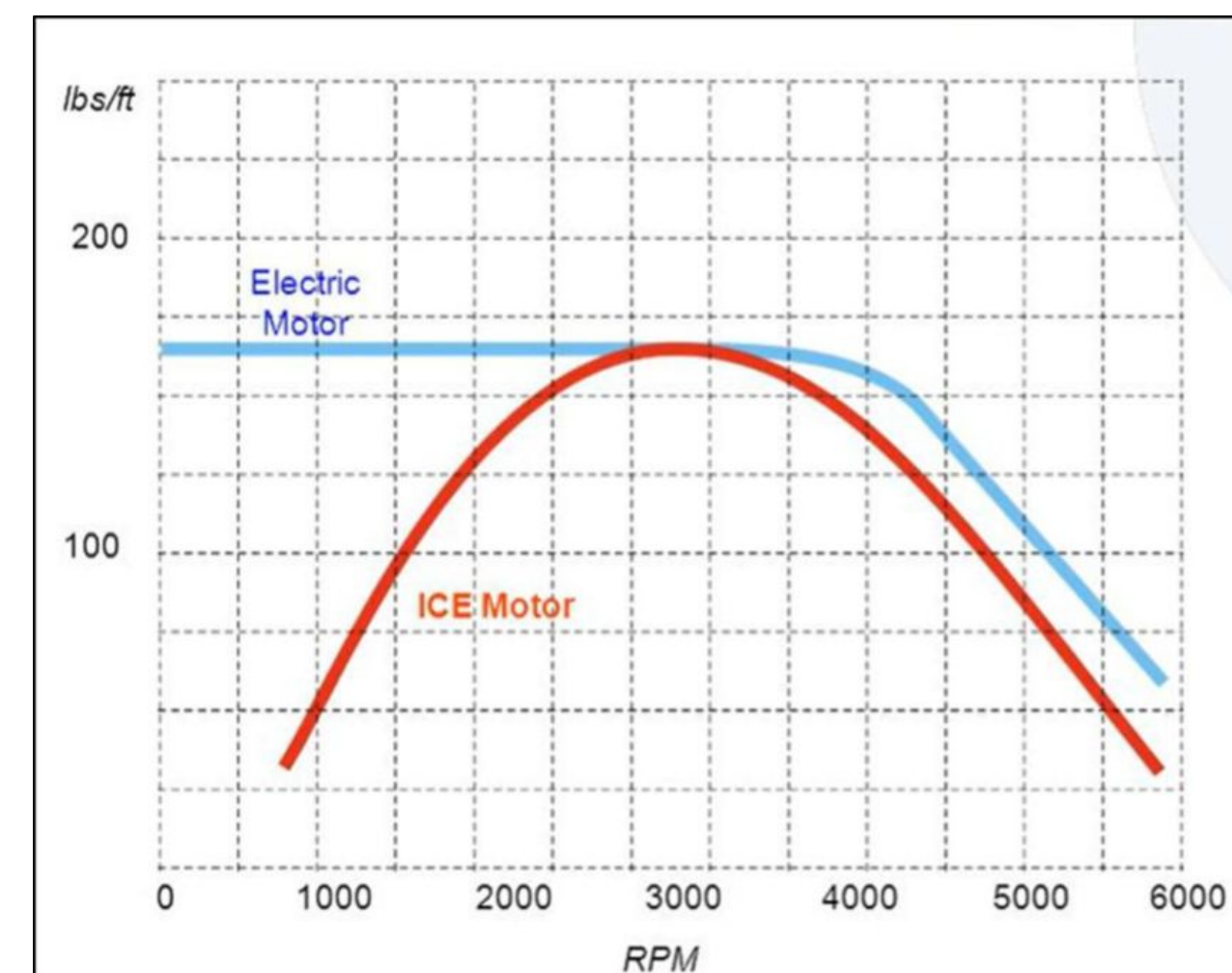
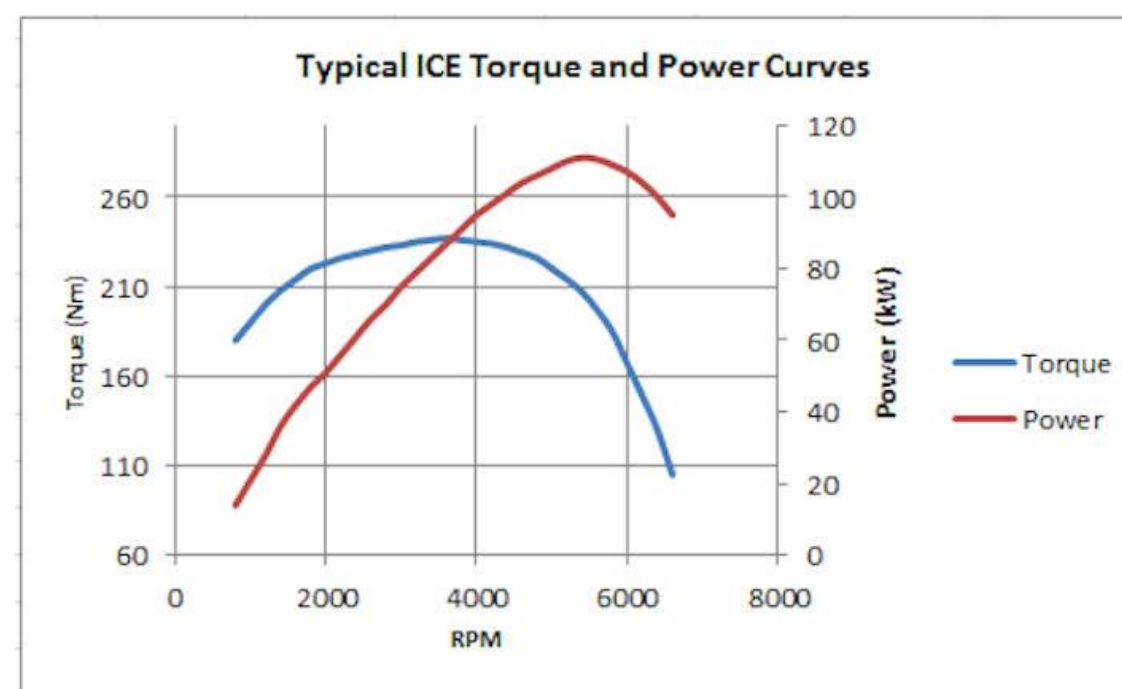
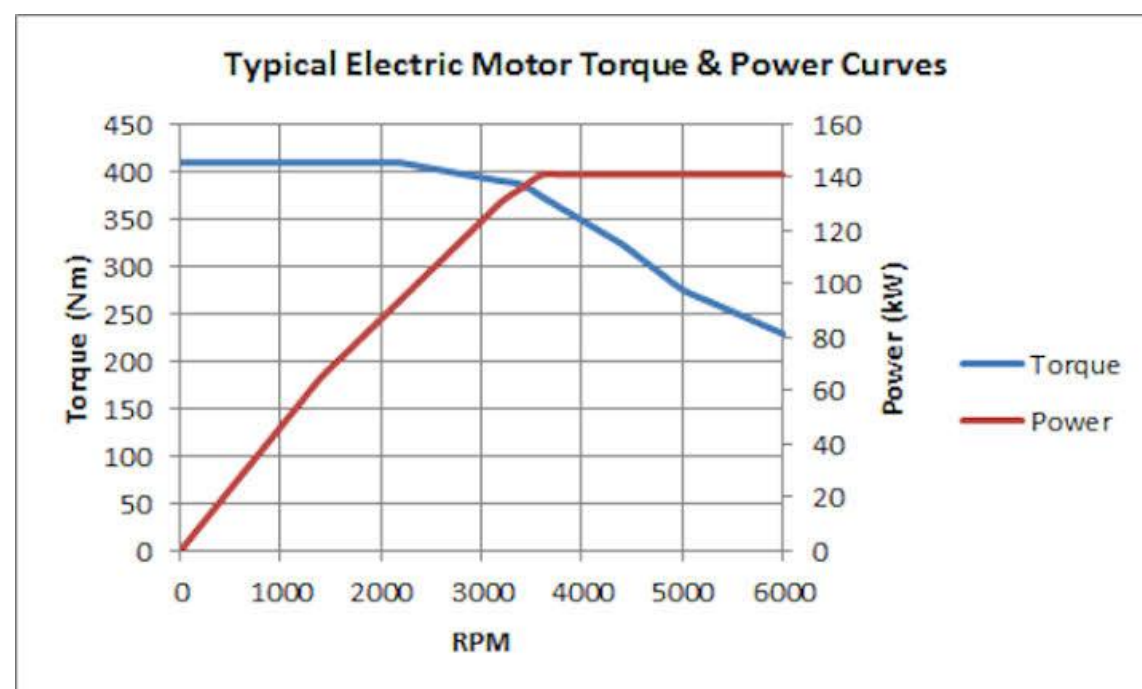
Electric Motor :

- Delivers instant torque from 0 RPM — eliminates need for a CVT
- Uses a planetary or fixed gearbox to reduce motor RPM to usable wheel speeds
- Simpler and more compact than IC setups
- Must account for motor over-speed protection at higher vehicle velocities
- Final chain drive still used for easy sprocket tuning



Comparing Both Types

Here you can clearly see the different trends going on in each graphs based on the type of engine used



Quick Quiz?

Answer: **C**

Reason:

Electric motors deliver instant torque from 0 RPM, so they don't need a CVT.

Instead, they use planetary or fixed gear reduction systems for compact, reliable power delivery.

Options A and B are false due to motor characteristics, and D is incorrect because electric motors can climb efficiently without shifting gears.

Which of the following is true for electric BAJA ATVs?

- A. They always use CVT
- B. They produce high torque only at high RPM
- C. They use planetary or fixed gear drives
- D. They cannot climb hills without gears

submit

Project Ideas

Dual Powertrain Design:

- Design a 2-speed gearbox adaptable for both petrol (IC) and electric motors.

Planetary Integration Project:

- Build a planetary gear reduction box and couple it with a BLDC motor for an eBAJA prototype.

CVT vs Fixed Gear Efficiency Test:

- Collect real-time data (RPM, speed, temp) and compare performance using data logging systems.

Sprocket Simulation in MATLAB:

- Write a script to simulate and visualize sprocket ratio changes, vehicle speed, and torque output.

Roadmap to Learning

1. Understand the Fundamentals:

- Study gear design basics, torque-speed curves, and power transmission principles.

2. Dive into CVT Mechanics:

- Learn about CVT operation, tuning, and types (Polaris/Nelson CVTs commonly used in BAJA).

3. Simulate and Analyze:

- Use MATLAB/Simulink to map torque curves and analyze how gear ratios affect performance.

4. Build a Working Drivetrain:

Resources...

Gear Design & Transmission Basics

- *Machine Design by R.S. Khurmi* – Gear theory, design standards
- [KHK Gear Manufacturer Design Guide](#) – Free online resource for practical gear design
- [Gear trains](#)

Simulation Tools

- [Simscape Driveline Tutorials – MathWorks](#)
- [MATLAB Drive – Torque-Speed Mapping Template](#)
- [ADAMS Vehicle Simulation Resources](#)

CVT and Chain Drive Systems

- [Understanding CVT](#)
- [Introduction to Chain Drives](#)

Electric Powertrain References

- [Electric powertrain Components](#)
- [Electric-Powertrain](#)

Career Prospects

Vehicle Dynamics Engineer

- Your BAJA experience with real-world tuning and testing gives you an edge in automotive R&D roles.

Drivetrain Developer

- Join top firms like Bosch, Mahindra Electric, or Continental working on next-gen transmission systems.

Simulation Expert

- Use tools like MATLAB, Simulink, Adams to simulate and optimize vehicle performance before prototyping.



EV Transmission Designer



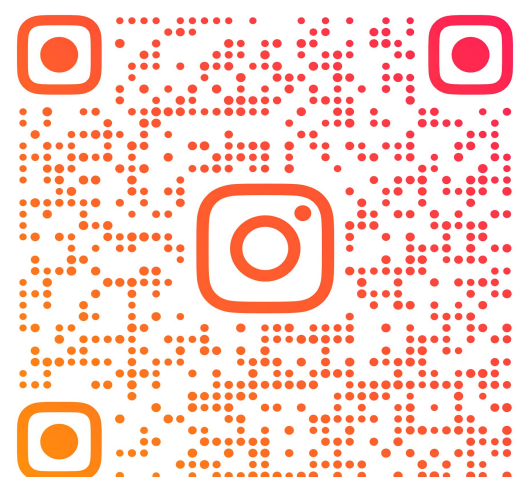
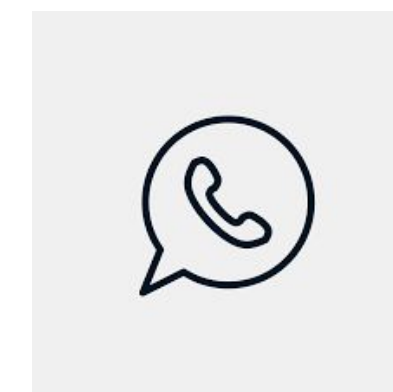
CONNECT WITH



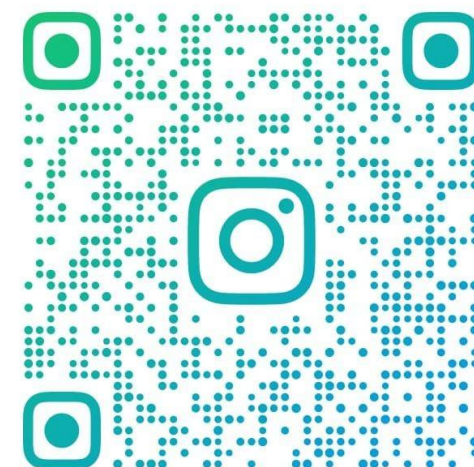
SAE CLUB
IIITDM KANCHEEPURAM



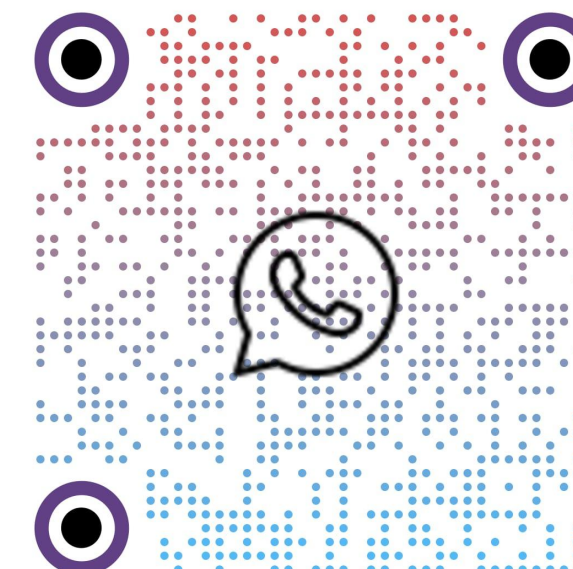
REVOLT RACERS



SAE_IIITDM



@REVOLT_RACERS_IIITDM





Thank You



SAE CLUB
IIITDM KANCHEEPURAM